

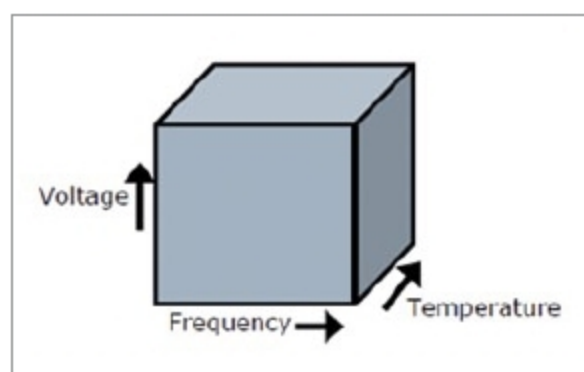


Fig. 7: Ideal operating range

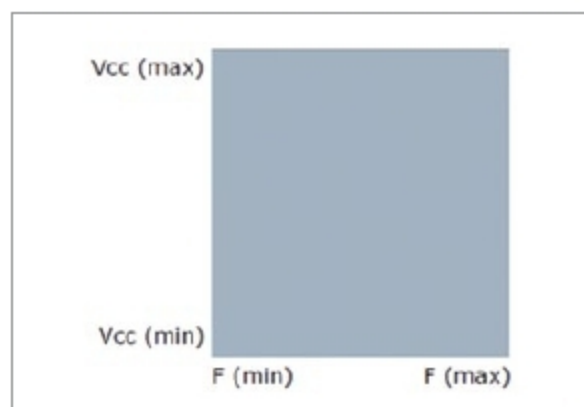


Fig. 8: 2D view of ideal case

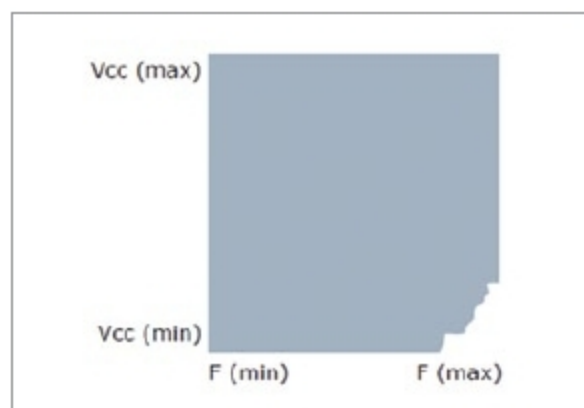


Fig. 9: 2D view of speed paths

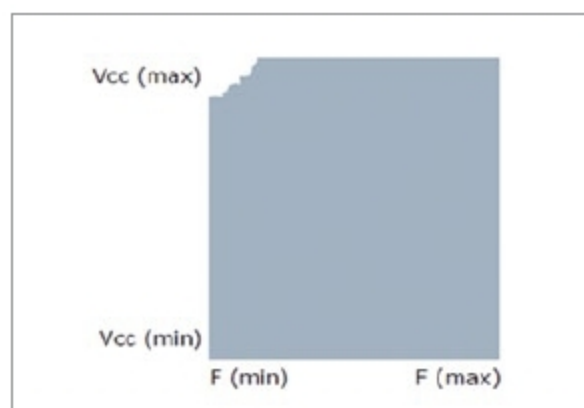


Fig. 10: 2D view of minimum delay

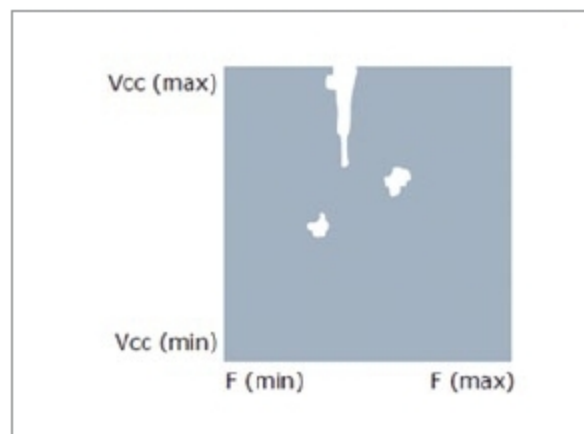


Fig. 11: 2D view of Shmoo holes/cracks

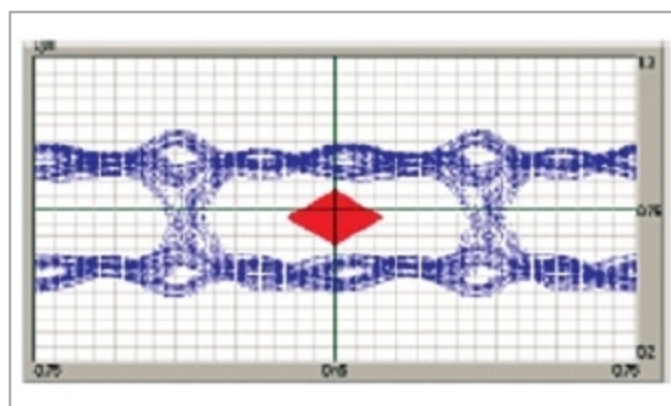


Fig. 12: Stimulus

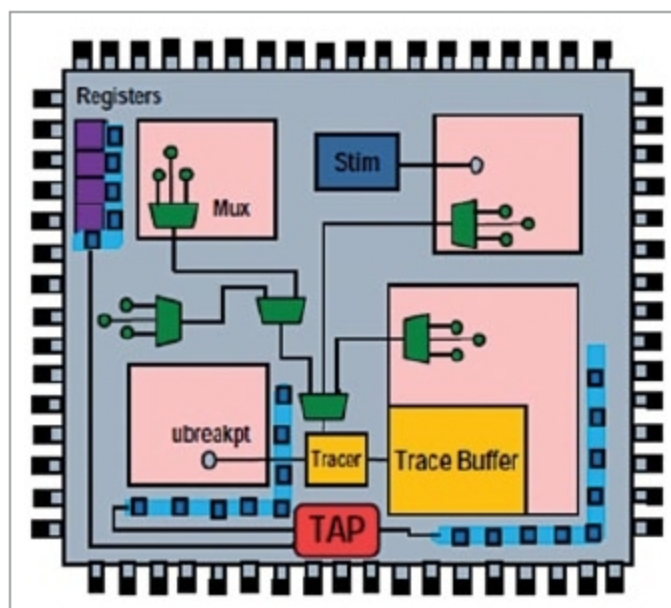


Fig. 13: Observability/control/survivability architecture

3. Fundamentals for circuit bug hunting are as follows:

- (a) Needs a sufficiently large population of devices
- (b) Needs to vary environmental conditions
- (c) Needs to stimulate stressful system behaviour
- (d) Stimulus is generally functional; failures look just like functional failures

Circuit debug issues

The following need to be checked for debugging:

1. On-die signal integrity
 - Cross-coupling induced noise
 - Droop-event induced noise
2. Power delivery integrity
 - High dynamic current events
 - Clock gating
3. Clock domain crossing
4. Process, voltage and temperature
 - Power state transitions
 - Silicon process variation

Ideal operating range

Ideally, silicon operates in well-defined volume. Minimum and maxi-

mum corners are defined as per manufacturers' specifications: uniform over-voltage, frequency, temperature, process and time. But what happens is a bit different.

Other factors include temperature, component age and silicon variability.

Speed paths

Circuit slows down as V_{cc} decreases. And, failure disappears as V_{cc} increases or F decreases. Historically, highest percentage of CPU circuit issues occur.

Minimum delays

Failure happens when circuit is too fast. Failure disappears as V_{cc} decreases or F increases. This is hard to fix.

Shmoo holes/cracks

1. Voids within the window
2. Intermittent working
3. Multiple clock domains
 - Skew within same domain
 - Skew/jitter across domains

Finding circuit margins

1. Exercise in platform-based silicon characterisation
2. Method is stress-to-fail (increase FMAX to failure)
3. Stimulus is directed/random
 - Victim/attacker patterns
 - Software load-driven power variation
 - Injected power state transitions
 - Randomised instructions, memory configurations and architectural events
4. Characterised before/after burn-in (simulate aging)
5. Characterised over large populations to understand silicon variability

I/O margin characteristics

1. Stimulus includes victim/attacker patterns, resonance stimulus and other noise generators like dynamic CPU core loads
2. V_{cc} and timing margined to fail (find extents of eye diagram)
3. Incorporates systematic 3D variation (shmooing) of voltage, temperature and frequency
4. Incorporates skewed silicon